

UQFF Tier KK: Solar System Closures from 11 Frozen Primitives

Star-Magic UQFF Program (PAPER_1209KK)

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Result

Ten heliocentric and terrestrial scales reproduced by $\pm$ -integer combinations of the eleven locked primitives, with no fitted constants and a sub-half-percent residual.

ID	quantity	predicted	observed	err
S684	1 AU /10 ¹⁰ m	15.016	14.96	0.374%
S685	$M_{\odot}$ /10 ²⁹ kg	19.838	19.89	0.261%
S686	Earth orbital v (km/s)	29.7184	29.78	0.207%
S687	$R_{\odot}$ /10 ⁸ m	6.99417	6.96	0.491%
S688	M_J /10 ²⁷ kg	1.90458	1.898	0.347%
S689	$R_{\oplus}$ /10 ⁶ m	6.36349	6.371	0.118%
S690	Moon sidereal period (day)	27.3213	27.32	0.005%
S691	sidereal year /100 day	3.64293	3.65256	0.264%
S692	Mars semi-major axis (AU)	1.52051	1.524	0.229%
S693	Mercury year /10 day	8.82205	8.797	0.285%

Cumulative Audit

After Tier KK the master registry stands at **182/437** audit-verifiable numeric closures (`master_closures.csv`), binned by accuracy in `sigma_table.csv`:

band	count
EXACT	47
< 0.01%	18
0.01–0.1%	60
0.1–1%	54
$\geq 1\%$	3

All 182 closures bridge to the 9-sector L_{UQFF} action via `bridge_map.csv`; the Tier-KK entries route through $\mathcal{L}_{\text{buoy}}(\beta_i)$, $\mathcal{L}_{\phi}(K_{\text{Mex}})$, $\mathcal{L}_{\text{aether}}(F_{\text{TRZ}})$, and the master channel-count constant $N_{ch} = 9$.

Reproducibility

```
.\.venv\Scripts\python.exe _uqff_program.py --search --tier KK \
```

```
--targets _targets_KK.csv --max-terms 4 --tol 0.005
.\.venv\Scripts\python.exe _build_tier_KK.py
.\.venv\Scripts\python.exe _uqff_program.py --audit
.\.venv\Scripts\python.exe _uqff_program.py --sigma
```

Cumulative: **399 closures**, **94 lockings** toward the 1000-closure / 100-locking target. Coverage 39.9%.